

UNIT V

19. Explain the construction, operation, V-I characteristics and application of SCR and explain its two transistor model.

Or

20. Briefly explain about the construction, operation, equivalent circuit V-I characteristics and application of UJT.
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B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Third Semester

Computer Science and Engineering

ELECTRONIC DEVICES AND CIRCUITS

(2013-14 Onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are clippers?
2. What is depletion region in PN junction?
3. Give some applications of BJT.
4. Why h parameter model is important for BJT?
5. Write notes on amplifier distortion.
6. Define frequency stability.
7. Mention any two characteristics of op-amp.

8. Specify the applications of op-amp.
9. Why SCR cannot be used as a bidirectional switch?
10. Give the advantages of LED.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the current flow in a forward biased PN junction with relevant expressions for minority carrier concentration. Also deduce the expression for total current density.

Or

12. Enumerate the concept of Zener breakdown in PN junction with its V-I characteristics.

UNIT II

13. Explain briefly about working of BJT emitter coupled differential amplifier.

Or

14. Explain the structure and operation of an N channel depletion type MOSFET with relevant characteristics.

UNIT III

15. Explain the construction and working of Hartley oscillator and derive the expression for frequency of oscillation.

Or

16. Briefly explain about the Colpitts oscillator. Also derive the expression for frequency of oscillation.

UNIT IV

17. With a neat diagram briefly explain about differential and common mode operation of op-amp.

Or

18. Write notes on :

- (a) Second order low pass filter.
- (b) Non-inverting amplifier.

UNIT V

19. With neat energy band diagrams, explain the V-I characteristics of Tunnel diode. Also discuss the negative resistance property of tunnel diode. (11)

Or

20. Explain the working of UJT as a relaxation oscillator with necessary wave forms and equations. (11)

5633029

B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Science and Engineering

ELECTRONIC DEVICES AND CIRCUITS

(2013 – 14 Onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define Avalanche break down.
2. What is peak inverse voltage?
3. What is frequency stability?
4. Mention the operating modes of MOSFET.
5. What are Push – Pull Circuits?
6. Define Harmonic Distortion.
7. Define Differential Inputs.

8. What is Common-Mode Rejection Ratio?
9. Write the working principle of light-emitting diode (LED).
10. What is TRIAC?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, One from each Unit.

All questions carry equal marks.

UNIT I

11. What do you understand about the depletion region at a PN junction, with the help of necessary diagrams and derive expression for barrier potential? (11)

Or

12. With neat sketches explain the operation of Zener Diode. (11)

UNIT II

13. Explain the construction and operation of a N-channel MOSFET in enhancement and depletion modes with the help of static drain characteristics and transfer characteristics. (11)

Or

14. With reference to a BJT, define the following terms and explain : (1)

- (a) Emitter efficiency
- (b) Base transportation factor
- (c) Large signal current gain. (1)

UNIT III

15. Draw and explain the operation of a Hartley oscillator. (1)

Or

16. Explain in detail about the operational characteristics of Series FED Class A amplifier. (11)

UNIT IV

17. Explain in detail about the basic characteristics of OP-AMP. (11)

Or

18. Explain in details about the practical OP-AMP circuits and its applications. (11)

16. (a) What is positive feedback in electronics? (3)
(b) Define Frequency stability of oscillators.
Give the factors affecting frequency stability. (4)
(c) Write the working of Wien Bridge oscillator. (4)

UNIT IV

17. (a) Write the characteristics of op-amp. (5)
(b) Give the applications of op-amp. (6)

Or

18. (a) Describe summing amplifier circuit. (4)
(b) Give the images of second order low pass and high pass active filters. (7)

UNIT V

19. (a) Define Seven segment display. (3)
(b) Explain the types of seven segment displays and controlling methods. (8)

Or

20. Write the operations and applications of TRIAC. (11)

5633029

B.Tech. DEGREE EXAMINATION, MAY 2019.

Third Semester

Computer Science and Engineering

ELECTRONIC DEVICES AND CIRCUITS

(2013–2014 onwards Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define clippers with a sample clipper circuit.
2. Draw and define voltage doubler and tripler circuits.
3. Give the emitter stabilized bias circuit.
4. Why do we need to bias a transistor?

5. What is power amplifier?
6. What is meant by feedback in electronics?
7. How does an operational amplifier work?
8. What is inverting op amp?
9. What is the use of tunnel diode?
10. What is DIAC? How many terminals are in DIAC?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

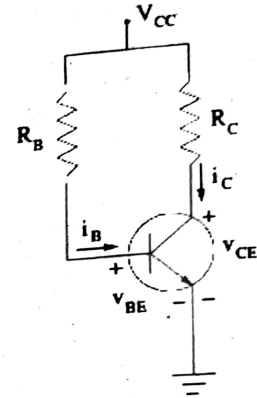
11. Explain construction, operation, characteristics and applications of pn junction diode.

Or

12. (a) Explain briefly about regulators and its types. (6)
- (b) Mention the applications of diode. (5)

UNIT II

13. Find values of R_C , R_B in the below circuit with $\beta = 100$ and $V_{CC} = 15$ V so that the Q-point is $I_C = 25$ mA and $V_{CE} = 7.5$ V.



Or

14. (a) Explain the small-signal models for the MOSFET. (5)
- (b) Discuss on the important parameters of BJT. (6)

UNIT III

15. What is Amplifier classes? Explain any two amplifier classes.

Or

5633029

B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.

Third Semester

Computer Science and Engineering

ELECTRONIC DEVICES AND CIRCUITS

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define ideal diode.
2. Annotate clippers and clampers.
3. State biasing.
4. Differentiate BJT and FET.
5. Write short notes on power amplifier and its types.
6. Define frequency stability.
7. List the characteristics of op-amp.

8. Enumerate the types of filters.
9. Differentiate DIAC and TRIAC.
10. State varactor diode.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the operation and characteristics of Zener diode.

Or

12. Explain series and shunt voltage regulators.

UNIT II

13. Explain the characteristics of BJT in Common Emitter Configuration.

Or

14. Discuss the construction and operation of MOSFET.

UNIT III

15. Describe the operation and frequency response of R-C coupled amplifier.

Or

16. Explain Wien bridge oscillators and its characteristics.

UNIT IV

17. Explain the differential and common mode operations of op-amp.

Or

18. Describe integrator and comparator operation of op-amp.

UNIT V

19. Explain the characteristics of UJT.

Or

20. Discuss the operation, characteristics and applications of SCR.

16. Explain about Class A transformer coupled amplifier and derive the expression for efficiency of the same.

UNIT IV

17. What are the stages of an operational amplifier? Explain them with a block diagram.

Or

18. (a) Design a circuit using operation amplifier that will compute the average of four input voltages. (5)
(b) Explain the concept of virtual ground. (6)

UNIT V

19. (a) Explain the operation of tunnel diode and draw its equivalent circuit. (6)
(b) Explain the operation of varactor diode. (5)

Or

20. (a) With a neat diagram, explain the operation and characteristics of a triac. (5)
(b) Explain the construction and operation of SCR with neat sketches and characteristics curves. (6)

5633029

B.Tech. DEGREE EXAMINATION, NOVEMBER 2019.

Third Semester

CSE

ELECTRONIC DEVICES AND CIRCUITS

(2013-14 onwards Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Write the current and junction voltage equation of a PN diode.
2. What is a zener shunt regulator?
3. Tabulate the differences between JFET and MOSFET.
4. Give the relationship between α , β and γ of a transistor.

5. Design a negative feedback amplifier that should have a bandwidth of 5 MHz, by finding the values of gain and feedback factor. Consider an amplifier with a bandwidth of 500 kHz and gain of 100.
6. Calculate the parameters of a Wien Bridge oscillator to operate at a frequency of 5 kHz.
7. Design a circuit using operational amplifier to perform integration of a given input signal.
8. List the characteristics of an operational amplifier.
9. How does an UJT differ from a BJT?
10. Write the application of LCD and LED.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. (a) Explain in detail about the switching characteristics on PN diode with a neat sketch. (6)
- (b) Draw the circuit of a clipper and explain. (5)

Or

12. There is an application which needs the output voltage to be regulated. Choose an appropriate diode/device, that would ensure this operation with appropriate circuit, describe how it regulates voltage. Consider a specific example, design the circuit with appropriate values of components involved. State the important constraints that need to be considered.

UNIT II

13. Why is the transistor so named? Explain with necessary diagrams, the construction and operation of a Bipolar Junction Transistor. Also discuss how BJT is used as : (a) a switch (b) an amplifier. Draw the characteristics curves of the BJT and explain.

Or

14. With neat diagrams explain the operation of MOSFET in Enhancement mode and derive its current equations.

UNIT III

15. A colpitt oscillator is designed with $C_1 = 100 \text{ pf}$ and $C_2 = 7500 \text{ pf}$. The inductance is variable. Determine the range of inductance values, if the frequency of oscillation is to vary between 950 KHz and 2050 KHz.

Or

5633029

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Third Semester

Computer Science and Engineering

ELECTRON DEVICES AND CIRCUITS

(2013-14 onwards batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Distinguish Avalanche break down and Zener breakdown.
2. What is barrier potential?
3. What are the advantages of FET over BJT?
4. Why BJT is called current controlled device?
5. State Barkhausen criterion for an oscillator.
6. Differentiate amplifier and oscillator.
7. Write the ideal characteristics of Op-Amp.

8. What is integrator?

9. Why SCR cannot be used as a bidirectional switch?

10. Give the symbol of TRIAC and DIAC.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the operation of forward and reverse characteristic of PN junction diode with circuit diagram and obtain its VI characteristic curve.

Or

12. Explain the series and shunt regulator using zener diode with the help of diagram.

UNIT II

13. Explain the voltage divides and fixed bias of BJT with the help of diagram.

Or

14. Explain the biasing techniques of MOSFET.

UNIT III

15. Explain the operation and frequency response RC coupled amplifiers with the diagram.

Or

16. Derive the expression for frequency characteristics of Wien bridge oscillator with help of diagram.

UNIT IV

17. Draw the internal block diagram of a typical Op-Amp and explain the function of each block.

Or

18. Write short notes on :

- (a) Differentiator
- (b) Comparator.

UNIT V

19. Explain the operation and VI characteristics SCR with the help of diagram.

Or

20. Explain the to principle of operation and characteristics of DIAC.

UNIT V

5633029

19. With neat energy band diagrams, explain the V-I characteristics of Tunnel diode. Also discuss the negative resistance property of tunnel diode. (11)

Or

20. Explain the working of UJT as a relaxation oscillator with necessary wave forms and equations. (11)

B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Science and Engineering

ELECTRONIC DEVICES AND CIRCUITS

(2013 – 14 Onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define Avalanche break down.
2. What is peak inverse voltage?
3. What is frequency stability?
4. Mention the operating modes of MOSFET.
5. What are Push – Pull Circuits?
6. Define Harmonic Distortion.
7. Define Differential Inputs.

8. What is Common-Mode Rejection Ratio?
9. Write the working principle of light-emitting diode (LED).
10. What is TRIAC?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, One from each Unit.

All questions carry equal marks.

UNIT I

11. What do you understand about the depletion region at a PN junction, with the help of necessary diagrams and derive expression for barrier potential? (11)

Or

12. With neat sketches explain the operation of Zener Diode. (11)

UNIT II

13. Explain the construction and operation of a N-channel MOSFET in enhancement and depletion modes with the help of static drain characteristics and transfer characteristics. (11)

Or

14. With reference to a BJT, define the following terms and explain :
 - (a) Emitter efficiency
 - (b) Base transportation factor
 - (c) Large signal current gain.

UNIT III

15. Draw and explain the operation of a Hartley oscillator. (11)

Or

16. Explain in detail about the operational characteristics of Series FED Class A amplifier. (11)

UNIT IV

17. Explain in detail about the basic characteristics of OP-AMP. (11)

Or

18. Explain in details about the practical OP-AMP circuits and its applications. (11)

5633029

B.Tech. DEGREE EXAMINATION, MAY 2018.

Third Semester

Computer Science and Engineering

ELECTRONIC DEVICES AND CIRCUITS

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. How transition capacitance differs from diffusion capacitance?
2. What do you meant by voltage doubles?
3. What are the advantages of PET amplifier over BJT?
4. Draw the small signal model of MOSFET.
5. Calculate the harmonic distortion components for an output signal having fundamental amplitude of 2.1 V, second harmonic amplitude of 0.3 V, third harmonic component of 0.1 V, and fourth harmonic component of 0.05 V.
6. Define conversion efficiency.

7. What are the ideal characteristics of an operation amplifier?
8. Calculate the CMRR (in db) for the circuit measurements of $V_d=1\text{ mV}$, $V_o=120\text{ mV}$, and $V_c=1\text{ mV}$, $V_o=20\mu\text{V}$.
9. Determine T1 for a varactor diode if $C_0=22\text{ pF}$, $\text{TCC}=0.02\%/^{\circ}\text{C}$ and $C=0.11\text{ pF}$ due to an increase in temperature above $T_0=25^{\circ}\text{C}$.
10. Sketch the symbol of DIAC and TRIAC.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. With suitable sketch, describe the I-V characteristics of a forward and reverse biased PN Junction diode and determine the expression for current from its basic principles.

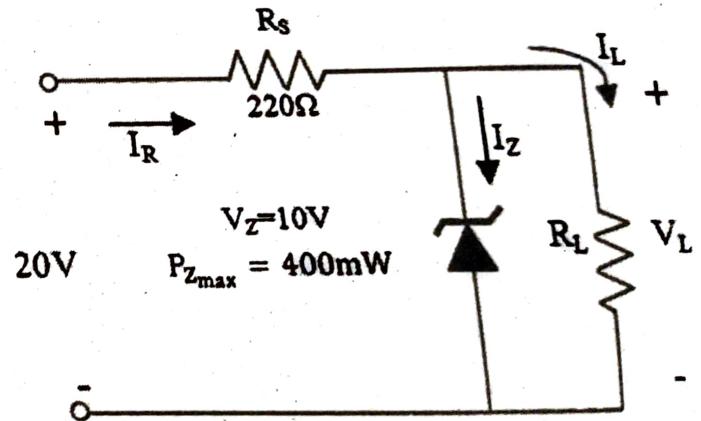
Or

- (a) Explain the basic principle of operation and characteristics of Uni-Junction Transistor with its equivalent circuit and diagram. (8)
- (b) Consider a Uni-Junction Transistor with $R_{B1} = 3k\Omega$ and $R_{B2} = 2k\Omega$. Calculate the intrinsic standoff ratio. (3)

12. (a) Determine V_L, I_L, I_z and I_R for the given network, if $R_L = 180\Omega$. (4)

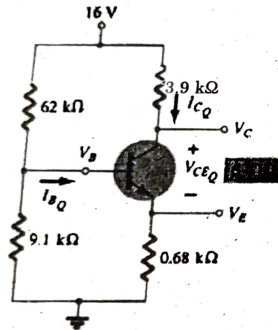
(b) Determine V_L, I_L, I_z and I_R for the given network, if $R_L = 330\Omega$. (4)

(c) Determine the value of R_L that will establish maximum power conditions for the Zener diode. (3)



UNIT II

13. For the voltage-divider bias configuration of the given circuit, determine I_{BQ} , I_{CQ} , V_{CEQ} , V_C , V_E and V_B .



Or

14. Draw the h-parameter equivalent circuit model of a bipolar transistor in a common-emitter circuit configuration. Derive the different parameters (A_V , A_I , Z_i and Z_o) involved in the model.

UNIT III

15. Explain a transformer coupled Class A b_1 Amplifier with neat diagram and obtain expression for efficiency.

Or

16. Derive the Expression for gain, input imped. and output impedance with feedback in vol shunt and current-series negative feed amplifiers.

UNIT IV

17. Explain operation of as three input non-inver summer, two input difference amp, differentiator and integrator by using OP-amp

Or

18. Explain the operation of second order low and high pass filters with neat diagrams.

UNIT V

19. Explain the characteristics of tunnel diode necessary energy band diagram and also me its applications.

Or

UNIT V

5633029

19. Explain the working principles of LED and LCD. Draw the truth table and circuit diagram of seven segment display.

Or

20. Describe the function, characteristics and their applications of DIAC and TRIAC.

B.Tech. DEGREE EXAMINATION,
APRIL/MAY 2017.

Third Semester

CSE

ELECTRONIC DEVICES AND CIRCUITS

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. How does to form a p-n junction diode?
2. What is clamping circuit?
3. What is meant by biasing of a transistor?
4. What is a JFET?
5. What is the power amplifier?
6. Mention two forms for amplifier distortion to occur.

UNIT III

7. List four characteristics of an ideal op-amp.
8. What is a comparator op-amp?
9. What is varactor diode?
10. What is intrinsic stand off ratio of UJT?

PART B — (5 × 11 = 55 marks)

Answer ALL questions ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Describe how does an ideal diode act as a switch. Mention any three applications of diode.

Or

12. Explain how does to describe AND, OR gates using diodes. Draw and explain the circuit diagram of voltage doubler and tripler.

UNIT II

13. How does JFETs differ from BJTs? Explain some Common FET Biasing Circuits.

Or

14. Describe small signal model of JFET and MOSFET with necessary circuit diagrams.

2

5633029

UNIT III

15. Define RC coupled amplifier. Explain the working principle and frequency response of RC-coupled amplifier. Also mention its advantages and disadvantages.

Or

16. What is the feedback of a electronic system? Why negative feedback is important in control systems? Give the conditions must be applied for oscillations to occur in a Wien Bridge Oscillator circuit

UNIT IV

17. Explain in detail with circuit diagram about the differential and common mode gain of differential amplifier.

Or

18. Describe and analyse differentiator and integrator with their circuit diagrams. Give applications of inverting and non-inverting op-amps.

3

5633029